

# THE STRUCTURAL SIEVE

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ABSTRACT. We take the prime base, not the interval, as the object of study, and measure the deterministic reach of its generative sieve. Existence of a prime beyond the base then follows as a consequence rather than as the goal: asking how far the base reaches turns the question into a single positivity statement about an exact window count, the window itself derived from the base, and we close that statement in the natural generality of the construction.

Starting from a complete prime base  $\mathcal{B} = \{q \text{ prime} : q < P_{\max}\}$  we separate two determinisms of the generative sieve. The base is *self-contained* exactly up to  $P_{\min} \cdot P_{\max}$ , where  $P_{\min}$  is the smallest prime that participates in the sieve: every composite in the window  $(P_{\max}, P_{\min}P_{\max}]$  is built entirely from the base, and immediately beyond the edge the first composite requiring a prime *outside* the base appears, namely  $P_{\min} \cdot \text{nextprime}(P_{\max})$ . Its *generative decision* (a survivor is void  $\Leftrightarrow$  prime) reaches further, throughout the zone  $(P_{\max}, P_{\max}^2)$ , and only there does it break. Inside the self-contained window this decision forces every survivor to be a new prime, so the base cannot cover its own window and a prime outside it is necessary; this forcing is the *atom*, and the self-contained edge is derived from the base rather than assumed. This yields a generalized window family  $(P_{\max}, P_{\min}P_{\max}]$  obtained by *truncated arithmetic* (sieving with a reduced base); its *tightest* member,  $P_{\min} = 2$ , is the window  $(P_{\max}, 2P_{\max}]$ , and every wider member follows from it by monotonicity. The whole construction is invariant under dilation of scale: the pair (window, base) is a single self-similar object, and this self-similarity is what the window count inherits.

The reduction sends non-emptiness of the window to a single positivity statement (the *atom*), which the proof settles in two regimes of different character. In the *constructive* regime the mechanism is generative: at every concrete scale it decides the window outright, since the sieve is finite and its evaluation decidable, so no counterexample can arise at any fixed step, and there the theorem is closed by structure alone. A generative process, however, cannot produce a single finite object for the completed infinite: the uniform statement over all scales at once has to be settled existentially rather than by construction. This is exactly where the central binomial coefficient  $\binom{2n}{n}$  supplies closure. Rather than serving as an external estimate, it functions as the object whose prime factorization records the window's new sources: a source  $q \in (n, 2n]$  divides  $\binom{2n}{n}$  exactly once. Its presence has nothing to do with any single fixed window: the proof needs to quantify over every window at once, and the coefficient enters for that reason, following from the paper's own purity law rather than being imported for the occasion. The extremal window ( $P_{\min} = 2$ ) is then closed by comparing the growth bound  $4^n \leq (2n + 1)\binom{2n}{n}$  with the old-source upper bound; both bounds are reproved within the development. These are the same two inequalities Erdős used in 1932, a convergence taken up in full in §10. As an immediate corollary, the extremal window recovers Chebyshev's strengthened form of Bertrand's 1845 postulate, a different question (existence of a prime, not the reach of a base) that happens to converge on the same object. The development is machine-verified in Lean 4 / Mathlib, and its main theorem does not rely on Mathlib's own formalization of Bertrand's postulate.

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## 1. INTRODUCTION

Bertrand's postulate asserts that for every integer  $n > 1$  there is a prime  $p$  with  $n < p \leq 2n$ . Bertrand conjectured it in 1845 on numerical evidence [1]; Chebyshev proved it in 1852 by analytic bounds on  $\vartheta$  and  $\psi$  [2]; Ramanujan gave a short  $\Gamma$ -function argument in 1919 [3]; and Erdős, in 1932, gave the elementary proof that has since become canonical, extracting the statement from divisibility properties of the central binomial coefficient  $\binom{2n}{n}$  [4]. Nair later observed that the same bounds follow from  $\text{lcm}(1, \dots, 2n)$  [5]. Each of these proofs stands on its predecessors; none discards the tools it inherits.

The present work stands on the same lineage, and says plainly where. Its contribution is a new *architecture* rather than a new inequality: Bertrand's postulate turns out to be the tightest instance of a self-similar family of window statements, the window and its right endpoint are *forced by the prime base* rather than posited, and the whole theorem localizes to a single positivity statement about an exact count. What is left at that one point is purely quantitative, and there we insert the lightest classical certificate, which turns out to be Erdős's pair of inequalities.

To the author's knowledge, and after comparing against the closest related material (Jacobsthal's function and its primordial-restricted gap bounds [6, 7], and the generalized Bertrand-type window results of [8]), three things set this construction apart; §10 records the comparison this claim rests on, and its limits. First, the window  $(P_{\max}, 2P_{\max}]$ , and more generally  $(P_{\max}, P_{\min}P_{\max}]$ , is *derived* together with its right endpoint from the base  $\mathcal{B} = \{q \text{ prime} : q < P_{\max}\}$  rather than posited: the endpoint is exactly the buildability edge of the base, self-containment fails provably one step beyond it, and the generative decision (void  $\Leftrightarrow$  prime) reaches all the way to the square of the anchor (§2, §3). Second, the statements assemble into a generalized family indexed by the smallest participating prime  $P_{\min}$ : Bertrand is its tightest member ( $P_{\min} = 2$ ), every wider member follows from it by monotonicity (§4), the family arises from *truncated arithmetic* (sieving with a reduced base), and it is manifestly invariant under dilation of scale (§5). Third, non-emptiness of the window collapses to a single exact-count statement, the atom, and whole regimes of it close with no counting at all (§6); what is left closes on the central binomial coefficient, whose prime factorization records the window's content (§7, §8).

A word on provenance belongs here too, if only briefly: this development does not simply repeat Erdős's argument under new names, nor does it claim to owe him nothing. §10 works that relationship out in full. For now it is enough to say that every claim below is machine-checked in Lean 4 with Mathlib, that the closing certificate is the only place a Mathlib fact about  $\binom{2n}{n}$  enters, and that Mathlib's own proof of Bertrand's postulate is never invoked.

Origin of the construction. The construction did not begin from Bertrand's postulate, and it is worth recording the observation from which it did begin, since that observation fixes the natural home of the statement. Consider the prime-factor incidence of the integers: for each  $n$ , record which primes divide it (Table 1). Read by rows, the primes are exactly the positions carrying no mark, the survivors of the factorization. Read by columns the table is more telling: a prime  $p$  makes its first appearance as a factor at  $2p$ , its first even multiple, and only from there does it begin to eliminate positions. Consequently the base of primes below a scale  $P$  already accounts for every composite up to  $2P$ : inside  $(P, 2P]$  every eliminated position is struck by a prime  $< P$ , so every survivor there is a new prime. What the observation measures is thus the *reach of deterministic factorization*: how far a fixed base generates the integers before a genuinely new prime is required. That reach is exactly  $2P$ . Whether the interval  $(P, 2P]$  happens to contain a prime was not the starting point; it is a corollary of the reach, read off afterwards.

Independence and naming. The right endpoint  $2P$ , the notion of a complete base, and the generative sieve built on them (§2–§6) were arrived at from this observation, independently of the classical postulate and of Erdős's proof; the coincidence of the tightest case with the interval of Bertrand's postulate was recognized only afterwards. We keep the classical

| $n$ | 2 | 3 | 5 | 7 | 11 | 13 | 17 | 19 | 23 |
|-----|---|---|---|---|----|----|----|----|----|
| 2   |   |   |   |   |    |    |    |    |    |
| 3   |   |   |   |   |    |    |    |    |    |
| 4   | • |   |   |   |    |    |    |    |    |
| 5   |   |   |   |   |    |    |    |    |    |
| 6   | • | • |   |   |    |    |    |    |    |
| 7   |   |   |   |   |    |    |    |    |    |
| 8   | • |   |   |   |    |    |    |    |    |
| 9   |   |   | • |   |    |    |    |    |    |
| 10  | • |   |   | • |    |    |    |    |    |
| 11  |   |   |   |   |    |    |    |    |    |
| 12  | • | • |   |   |    |    |    |    |    |
| 13  |   |   |   |   |    |    |    |    |    |
| 14  | • |   |   |   | •  |    |    |    |    |
| 15  |   |   | • | • |    |    |    |    |    |
| 16  | • |   |   |   |    |    |    |    |    |
| 17  |   |   |   |   |    |    |    |    |    |
| 18  | • | • |   |   |    |    |    |    |    |
| 19  |   |   |   |   |    |    |    |    |    |
| 20  | • |   |   | • |    |    |    |    |    |
| 21  |   |   | • |   | •  |    |    |    |    |
| 22  | • |   |   |   |    |    |    |    |    |
| 23  |   |   |   |   |    |    |    |    |    |
| 24  | • | • |   |   |    |    |    |    |    |

TABLE 1. Prime-factor incidence for  $2 \leq n \leq 24$ . A mark in column  $p$  records  $p \mid n$ ; the empty rows ( $n = 2, 3, 5, 7, 11, 13, 17, 19, 23$ ) are the primes, the survivors of the factorization. Each prime  $p$  first appears as a factor at  $2p$ , so the primes  $\geq 13$  carry no mark below  $2 \cdot 13 = 26$ : the rule at  $P_{\max} = 11$  separates the emerged base from the primes not yet required. The base of primes  $< P$  thus covers every composite in  $(P, 2P]$ , and the reach of the fixed base is exactly  $2P$ , the endpoint derived in §3.

name by historical priority, while noting that the statement here is naturally one about *primes* rather than natural numbers: the object is the window  $(P_{\max}, P_{\min}P_{\max}]$  anchored at a prime scale, not  $(n, 2n]$  anchored at an arbitrary integer, and the two coincide precisely at the tightest member  $P_{\min} = 2$  (Corollary 9.3).

Throughout,  $P$  denotes a prime playing the role of the current scale (the anchor  $P_{\max}$ );  $\mathcal{B}(P) = \{q \text{ prime} : q < P\}$  is the base below it; and, for a finite set  $S$  of primes, a position  $n$  is *S-void* if no element of  $S$  divides  $n$ . All Lean identifiers are typeset as `like_this` and refer to the accompanying formalization,<sup>1</sup> in which each may be located by name.

## 2. THE PRIME BASE AND ITS DETERMINISTIC REACH

**Definition 2.1** (Base and void). For a prime  $P$  let the *complete base below  $P$*  be  $\mathcal{B}(P) := \{q \text{ prime} : q < P\}$ , and let the *Bertrand window* be  $W(P) := (P, 2P] \cap \mathbb{Z}$ . A position  $n$  is  *$\mathcal{B}(P)$ -void* if  $q \nmid n$  for every  $q \in \mathcal{B}(P)$ , equivalently if  $\gcd(n, \prod_{q < P} q) = 1$ .

*Formalization.* `isVoid`, `isVoid_iff_coprime`, `gps_window`;  $\prod_{q < P} q$  is `primorial_below`.

**Remark 2.2** (One scale parameter fixes base, window, and width). A single prime  $P$  determines all three data: it is the largest prime of the base (the complete set of primes up to  $P$ , with none omitted), the left endpoint of the window  $W(P) = (P, 2P]$ , and its width  $|W(P)| = P$ . The base and the window are thus both functions of  $P$ , not independent

<sup>1</sup>Publicly available at <https://github.com/Flaman55/StructuralSieve>.

parameters. (Whether  $P$  itself is counted in the base is immaterial to the covering: its only multiple in the window is  $2P$ , already divisible by 2; see `anchor_idle_in_own_window`.) The same structure underlies Bertrand’s classical window  $(n, 2n]$ , with  $n$  in the role of  $P$ : the present statement is phrased on a prime scale rather than an arbitrary integer, and the two coincide at the extremal member  $P_{\min} = 2$  (§9). The quantity that governs the statement is accordingly the local deficiency of the base on its own window (the length of the longest interval of covered positions in  $W(P)$ ), which external gap functions only bound from above.

The one arithmetic fact that governs the window is the least-prime-factor bound: a composite that is not too large must reveal a small factor.

**Lemma 2.3** (Least prime factor in the window). *Let  $2 < P$  and  $n \in W(P)$ . If  $n$  is composite then its least prime factor satisfies  $\text{minFac}(n) \leq \sqrt{2P} < P$ ; in particular  $\text{minFac}(n) \in \mathcal{B}(P)$ .*

*Proof.* If  $n = ab$  with  $1 < a \leq b$  then  $a \leq \sqrt{n}$ . For  $n \in W(P)$  we have  $n \leq 2P$ , so the least prime factor is  $\leq \sqrt{2P}$ , and  $\sqrt{2P} < P$  for  $P > 2$ .  $\square$

*Formalization.* `least_prime_factor_bound` (`LPF.lean`), `window_composite_minFac_sq_le` (`Rings.lean`).

Lemma 2.3 turns void into primality inside the window: if a position survives the base, it has no factor below  $P$  and is not composite, hence prime.

**Theorem 2.4** (Determinism of the window). *Let  $P$  be prime with  $2 < P$  and  $n \in W(P)$ . Then  $n$  is  $\mathcal{B}(P)$ -void if and only if  $n$  is prime.*

*Proof.* If  $n$  is prime and  $n > P$  then no  $q < P$  divides  $n$ , so  $n$  is void. Conversely, a void  $n$  has no prime factor below  $P$ ; were it composite, Lemma 2.3 would supply one. Hence  $n$  is prime.  $\square$

*Formalization.* `void_in_window_prime`, `void_iff_prime_in_window`, `void_iff_prime_in_deterministic_zone`; the underlying “composite  $\Rightarrow$  covered by an earlier prime” step is `composites_covered_by_prev` and `prime_iff_uncovered_by_prev` (`ZeroForce.lean`).

Two consequences frame the rest of the paper. First, non-emptiness of  $W(P)$  as a set of primes is *equivalent* to the existence of a single  $\mathcal{B}(P)$ -void position; this is the object we will count. Second, the mechanism of Theorem 2.4 does not depend on  $P$ : it has the same form at every anchor.

**Proposition 2.5** (Scale invariance of the mechanism). *For every prime  $P$  with  $2 < P$  and every  $n \in W(P)$ , the equivalence  $(n \text{ is } \mathcal{B}(P)\text{-void}) \Leftrightarrow (n \text{ is prime})$  holds verbatim. The rule is invariant under the choice of anchor.*

*Formalization.* `mechanism_scale_invariant` (`Rings.lean`).

Proposition 2.5 is the precise content of the informal statement that the construction “behaves the same at the start of the number line and at any later point.” We return to it in §4: the pair  $(W(P), \mathcal{B}(P))$  is a single object viewed at scale  $P$ , and moving along  $\mathbb{N}$  rescales window and base together.

### 3. THE BUILDABILITY EDGE: WHY $P_{\min}P_{\max}$

The window’s right endpoint is not chosen; it is the exact reach of the base. Fix the smallest prime allowed to participate in the sieve,  $P_{\min}$  (for Bertrand,  $P_{\min} = 2$ ), and a scale  $P_{\max}$ .

**Theorem 3.1** (Buildability). *Let  $2 \leq P_{\min}$  and let  $n$  be composite with  $\text{minFac}(n) \geq P_{\min}$  and  $n \leq P_{\min}P_{\max}$ . Then every prime factor of  $n$  is  $\leq P_{\max}$ ; that is,  $n$  is fully built from primes in  $[P_{\min}, P_{\max}]$ .*

*Proof.* Let  $r$  be any prime factor of  $n$  and write  $n = r s$ . Since  $n$  is composite,  $s \geq \text{minFac}(n) \geq P_{\min}$ , hence  $r = n/s \leq P_{\min} P_{\max} / P_{\min} = P_{\max}$ .  $\square$

*Formalization.* `rough_composite_smooth` (general  $P_{\min}$ ), `window_composite_smooth` (case  $P_{\min} = 2$ ).

Thus below  $P_{\min} P_{\max}$  every composite with least factor  $\geq P_{\min}$  is  $P_{\max}$ -smooth: the base can build it, and a surviving position can only be a new prime. The edge is sharp.

**Proposition 3.2** (First unbuildable composite). *The smallest composite with least factor  $\geq P_{\min}$  that is not  $P_{\max}$ -smooth is  $P_{\min} \cdot \text{nextprime}(P_{\max})$ , which exceeds  $P_{\min} P_{\max}$ . Hence the reach  $P_{\min} P_{\max}$  is exact.*

For  $P_{\min} = 2$  the self-contained reach is  $2P_{\max}$ . Two distinct boundaries meet the base here and must not be conflated. The base's *self-containment* ends at  $2P$ : just beyond it a composite requiring a prime outside the base appears (Proposition 3.2), and  $2P$  is the largest interval above  $P$  in which no element properly divides another (Theorem 3.4); this is what fixes the window. The sieve's *generative decision* (the equivalence  $\text{void} \Leftrightarrow \text{prime}$  of Theorem 2.4) reaches much further, throughout the zone  $(P, P^2)$ , and breaks only at the square of the anchor.

**Theorem 3.3** (Generative decision reaches the square scale). *Let  $P$  be prime,  $2 < P$ . Every  $\mathcal{B}(P)$ -void  $n$  with  $P < n < P^2$  is prime, and this fails at  $n = P^2$ : the square  $P^2$  is composite yet  $\mathcal{B}(P)$ -void. Hence the generative decision of Theorem 2.4 reaches exactly the zone  $(P, P^2)$  (the square of the anchor), far beyond the self-contained edge  $2P$ .*

*Proof.* If  $P < n < P^2$  is  $\mathcal{B}(P)$ -void then no prime  $< P$  divides it, so every prime factor of  $n$  is  $\geq P$ ; were  $n$  composite it would be a product of two such factors, hence  $\geq P^2$ , against  $n < P^2$ . So  $n$  is prime. For the failure,  $P^2$  is composite and its only prime factor is  $P \notin \mathcal{B}(P)$ , so it is  $\mathcal{B}(P)$ -void; the least prime  $q > P$  gives a further composite void  $q^2 > P^2$ .  $\square$

*Formalization.* Positive zone `void_iff_prime_in_deterministic_zone`; the break is witnessed in `determinism_breaks_above` (`Rings.lean`), whose exhibited composite void is  $q^2$ .

It is self-containment, not the generative decision, that isolates *why*  $2P$  is the window's true boundary:  $(P, 2P]$  is exactly the largest interval above  $P$  in which no element properly divides another, so elimination can proceed only through the base and never through a survivor.

**Theorem 3.4** (Self-containment, sharp at  $2P$ ). *For  $m, n \in W(P)$ ,  $m \mid n$  implies  $m = n$ : no proper divisibility holds inside  $W(P)$ . This fails immediately above the edge:  $P + 1 \in W(P)$  but its proper multiple  $2(P + 1) > 2P$  leaves the window.*

*Proof.* If  $m \mid n$  with  $m, n \in (P, 2P]$  and  $m \neq n$ , then  $n \geq 2m > 2P$ , contradicting  $n \leq 2P$ . The element  $2(P + 1)$  is the first proper multiple of an in-window element to exceed  $2P$ .  $\square$

*Formalization.* `window_self_contained_dvd`, `self_containment_breaks_just_above` (`Rings.lean`); `window_self_contained_bound`, `max_self_contained_width` (`SelfContained.lean`).

Self-containment holds at every scale, not only for large bases; a small window contains no proper multiple simply because it is too narrow, not because the property fails there. §4 shows this structural property belongs specifically to the tightest window, present exactly where the family is most constrained.

## 4. THE GENERALIZED WINDOW FAMILY

**Definition 4.1** (Generalized window). For  $P_{\min}, P_{\max}$  let  $\text{gW}(P_{\min}, P_{\max}) := (P_{\max}, P_{\min}P_{\max}]$  and say it *has a void* if some  $n \in \text{gW}(P_{\min}, P_{\max})$  is  $\mathcal{B}(P_{\max})$ -void.

*Formalization.* `gwindow, gwindowHasVoid (Rings.lean)`.

**Proposition 4.2** (Bertrand is the case  $P_{\min} = 2$ ).  $\text{gW}(2, P_{\max}) = W(P_{\max})$ , and  $\text{gW}(2, P_{\max})$  has a void iff  $W(P_{\max})$  contains a prime.

*Formalization.* `gwindow_two_eq_window, gwindowHasVoid_two_iff`.

The family is monotone in the width parameter: widening the window can only help.

**Theorem 4.3** (Monotonicity: tight  $\Rightarrow$  wide). If  $P_{\min} \leq P'_{\min}$  then  $\text{gW}(P_{\min}, P_{\max}) \subseteq \text{gW}(P'_{\min}, P_{\max})$ ; hence a void for the narrower window is a void for the wider one. In particular every member of the family follows from the tightest member  $P_{\min} = 2$ .

*Proof.*  $P_{\min}P_{\max} \leq P'_{\min}P_{\max}$  gives the inclusion of half-open intervals with the same left endpoint; a void position and its voidness are preserved.  $\square$

*Formalization.* `gwindow_subset, gwindowHasVoid_mono, gwindowHasVoid_of_bertrand`.

At the wide end the statement is free: once  $P_{\min}P_{\max}$  is large enough, a prime in the window is delivered by elementary means (a Euclid-type witness), with no counting.

*Formalization.* `gwindowHasVoid_of_wide`.

Two points are worth spelling out. By Theorem 4.3 the narrow case implies the wide ones, so the family carries no more content than Bertrand itself: it is *equivalent* to it, not strictly stronger, and everything is concentrated at  $P_{\min} = 2$ . The structure is also richest exactly there: self-containment (Theorem 3.4) holds *only* at  $P_{\min} = 2$ , since any wider window already contains  $2(P_{\max} + 1)$ , a proper multiple of an in-window element. So the tightest window carries the most structure, which is also why it is the hardest case to establish.

This is the place to state the scale invariance the family embodies. The invariant is not a reflection of coordinates but a dilation: the object is the pair  $((X, 2X], \{q \text{ prime} : q < X\})$ , and under  $X \mapsto \lambda X$  the window and the base rescale *together*, the ratio 2 being preserved. Freezing the base at one scale while sliding the window to another does not even belong to the family; it mismatches the two dilations. Bertrand's  $(n, 2n]$  is simply the unit cell of this scale,  $P_{\min} = 2$ ; motion along  $\mathbb{N}$  is the renormalization  $n \mapsto \lambda n$  with window and base co-scaling. Proposition 2.5 is the invariance of the local rule; the count that follows inherits it exactly (§6, `coverFiber_eq_scaled`).

## 5. TRUNCATED ARITHMETIC: THE REDUCED-BASE SIEVE

The family of §4 is produced by *truncated arithmetic*: instead of the full base one sieves with the reduced base of primes that can actually reach the window. By Lemma 2.3 the only rings that cover a composite  $n \leq 2P$  are those with  $p \leq \sqrt{2P}$ ; equivalently  $p^2 \leq 2P$ . This selects a truncated primorial.

**Definition 5.1** (Truncated primorial).  $M_{\text{tr}}(P) := \prod_{\substack{p < P \text{ prime} \\ p^2 \leq 2P}} p$ .

*Formalization.* `truncPrimorial`; general version `gtruncPrimorial`.

**Theorem 5.2** (Void through the truncated base). For  $2 < P$  and  $n \in W(P)$ ,  $n$  is  $\mathcal{B}(P)$ -void iff  $\gcd(n, M_{\text{tr}}(P)) = 1$ . Consequently  $W(P)$  has a void iff  $(P, 2P]$  contains an integer coprime to  $M_{\text{tr}}(P)$ .

*Proof.* Coprimality to the full base clearly implies coprimality to the truncated base. Conversely, if  $n$  is coprime to  $M_{\text{tr}}(P)$  but divisible by some prime  $p < P$ , then  $n$  is composite, so by Lemma 2.3  $\text{minFac}(n)^2 \leq 2P$ , and  $\text{minFac}(n)$  divides  $M_{\text{tr}}(P)$ : a contradiction.  $\square$

*Formalization.* `void_iff_coprime_trunc`, `windowHasVoid_iff_trunc_coprime`, `survivor_of_trunc_coprime`; general `gvoid_iff_coprime_trunc`, `gwindowHasVoid_iff_trunc_coprime`.

The truncation principle makes the reach explicit at every level and exposes the self-similarity. Removing only the prime 2 leaves reach  $3n$ ; removing 2 and 3 leaves  $5n$ ; in general the smallest omitted prime dictates the reach. The buildability map is therefore layered: the class of positions with least factor  $p$  is buildable up to  $pP_{\text{max}}$ , and the same phenomenon recurs at each level with Bertrand as the base case  $P_{\text{min}} = 2$ .

*Formalization.* Reduced-base sieve and its non-closure: `exists_uncovered_of_card_lt`, `truncated_not_sieve_closed`, `card_truncated_window` (`Truncated.lean`).

## 6. REDUCTION TO A SINGLE WINDOW-COUNT ATOM

We now count. Let  $\text{cov}(P)$  be the number of composite (covered) positions in  $W(P)$ ; since  $|W(P)| = P$ , non-emptiness is a strict inequality.

**Definition 6.1** (Covered count).  $\text{cov}(P) := \#\{n \in W(P) : n \text{ is not } \mathcal{B}(P)\text{-void}\}.$

**Theorem 6.2** (Reduction to an atom).  *$W(P)$  has a void iff  $\text{cov}(P) < P$ . Hence Bertrand at scale  $P$  is equivalent to the single positivity statement  $P - \text{cov}(P) > 0$ .*

*Formalization.* `coveredCount`, `gps_window_card`, `windowHasVoid_iff_coveredCount_lt`, `coveredCount_lt_imp_nonempty`.

A structural preliminary records that the largest ring  $P_{\text{max}}$  exerts no force inside its own window (its first multiple  $2P_{\text{max}}$  is the lone edge point), and that the previous base carries a weight  $w = P_{\text{max}} \cdot \varphi(M)/M \geq 1$ , proved once and for all, independently of scale. This weight is structural content about the setting, not a hypothesis either regime's closure consumes below: the sparse regime closes by a union bound (Theorem 6.5 and the paragraph preceding it), and the dense regime closes on the central binomial coefficient (§8), neither appealing to  $w \geq 1$ . It is recorded here as a by-product of the reduction, not because the atom's proof term calls on it.

*Formalization.* `zero_effective_force` (`ZeroForce.lean`); weight  $w \geq 1$ : `WeightGeOne`, `weight_base`, `weight_step`, `structural_weight_ge_one`, `prev_base_cannot_cover_window` (`Weight.lean`); bridge `weight_density_bridge`; anchor `idle_anchor_idle_in_own_window`. This chain is proved and wired through `structural_sieve_survivor`'s signature, but `dense_sieve_survivor` (the lemma that actually closes the dense regime) takes the resulting density fact as an unused parameter (`_hdensity`) and proves the window nonempty directly from `binomial_contradiction` instead.

The covered count decomposes disjointly by least prime factor, and each fiber is, by an exact bijection, the *same* coprimality count one scale down, the recursion engine promised by scale invariance.

**Theorem 6.3** (Disjoint minFac decomposition and self-similar fibers).  $\text{cov}(P) = \sum_p A_p(P),$

where  $A_p(P)$  counts the  $n \in W(P)$  with  $\text{minFac}(n) = p$ , the sum ranging over primes with  $p^2 \leq 2P$ . Moreover

$$A_p(P) = \#\left\{m \in \left(\frac{P}{p}, \frac{2P}{p}\right] : \gcd(m, \prod_{r < p} r) = 1\right\},$$

the same window-coprimality problem at scale  $P/p$ .

*Formalization.* `coveredCount_eq_sum_minFac_fiber`, `coveredCount_eq_sum_truncated`, `coverFiber_eq_scaled`; the dominant ring  $p = 2$  is split off exactly in `coveredCount_split_two`, `windowHasVoid_iff_oddSum_lt`.

Two exact handles on  $\text{cov}(P)$  follow. A union bound with multiplicity gives  $\text{cov}(P) \leq \sigma(P) := \sum_p (\text{hits of } p)$ ; the gap between the distinct count  $\text{cov}(P)$  and  $\sigma(P)$  is exactly the overlap (redundancy) between rings, so  $\sigma(P) > P$  does not obstruct  $\text{cov}(P) < P$ . The full, signed identity is inclusion–exclusion (Legendre).

**Theorem 6.4** (Interference identity). *For a finite set  $S$  of primes and  $a \leq b$ ,*

$$\#\{n \in (a, b] : \gcd(n, \prod_{q \in S} q) = 1\} = \sum_{T \subseteq S} (-1)^{|T|} \left( \lfloor \frac{b}{\prod_{q \in T} q} \rfloor - \lfloor \frac{a}{\prod_{q \in T} q} \rfloor \right).$$

*The order-by-order layers are its partial sums; the whole is an exact identity.*

*Formalization.* `coveredCount_le_sigma`, `interference_step`, `interference_formula` (`Rings.lean`).

Entire regimes of the atom close with no analysis. In the *sparse* regime a single union bound produces a survivor. For small anchors a sufficient condition settles the window: only the *active-covering rings*  $p^2 \leq 2P$  can cover a composite in  $W(P)$  (Lemma 2.3), so, writing  $M_{\text{tr}}(P) = \prod_{p^2 \leq 2P} p$ , if every run of  $P$  consecutive integers meets a number coprime to  $M_{\text{tr}}(P)$  (i.e. the coprime-gap  $g(M_{\text{tr}}(P)) \leq P$ ), then  $W(P)$  contains a coprime, hence a void.

**Theorem 6.5** (Small-anchor closure). *If  $g(M_{\text{tr}}(P)) \leq P$  then  $W(P)$  has a void. Concretely  $g(30) = 6$ ,  $g(210) = 10$ ,  $g(2310) = 14$  close all anchors  $P \leq 83$  unconditionally.*

A caveat on this bound. Here  $g(M_{\text{tr}}(P))$  is the global coprime-gap of a *fixed* modulus: the product of the active-covering rings, a subcollection that coincides with a smaller, foreign base, not the complete base of the object. It only *upper-bounds* the object’s own quantity, the local run of covered positions inside  $W(P)$ ; the inequality  $g \leq P$  is therefore a loose sufficient proxy, not the governing structure of the object, and it ceases to be available at large  $P$ , where the residual dense regime is settled by the object below, not by any gap function.

*Formalization.* `windowHasVoid_of_jacobsthal`, `windowHasVoid_of_trunc_gap`, `jacobsthal_thirty`, `jacobsthal_210`, `jacobsthal_2310`, and `windowHasVoid_3 ... windowHasVoid_83` (`Rings.lean`).

This closure is independent of the assembled proof, not a stage feeding into it: `structural_sieve_survivor` never branches on  $P \leq 83$ , and dispatches every prime  $P$  through the sparse/dense split of §6, the dense side closed uniformly by `binomial_contradiction` at the single threshold  $n = 512$  (§8). Theorem 6.5 thus supplies a second, self-contained certificate for these finitely many scales, disjoint in its derivation from Theorem 9.1; it is retained to exhibit a regime that closes with no counting at all, not because the main theorem’s proof term invokes it.

What remains is the *dense* regime at large  $P$ , where the structural mechanism has both a definite reach and a definite limit. At every *concrete* scale  $P$  the construction decides the window outright: the sieve is finite and deterministic, each finite step is verified, and no counterexample can arise, so the statement “ $W(P)$  has a void” is decidable at every fixed  $P$ . Generation alone, though, cannot deliver the *uniform* statement over all scales at once: reaching “for every  $P$ ” that way would mean carrying the construction to infinity, and no finite computation does that. Going from “true at every finite step” to “true in the infinite” is an existential step, not a computational one, and we make no claim to close it generatively; none is known. The central binomial coefficient is what closes this infinite tail, the part beyond any finite computation, and the next section explains why it is the right object for the job.



*Formalization.* Regime dispatch: `structural_sieve_survivor`; the residual dense case `dense_sieve_survivor` (`GPS_StateMachine.lean`).

## 7. THE CENTRAL BINOMIAL COEFFICIENT AND THE WINDOW'S PRIME CONTENT

Since the dense regime closes existentially, the central binomial coefficient is what does the closing. Its only job is to settle the infinite tail that no finite generation reaches: a single uniform argument standing in for a process that cannot be run to completion, not a fix for the finite mechanism, which already decides every concrete scale on its own.

That  $\binom{2n}{n}$  is the appropriate object here is itself structural. It is an entry of Newton's binomial expansion:  $4^n = (1+1)^{2n}$  is the sum of the  $2n+1$  entries of the  $2n$ -th Pascal row, of which the central one is largest, and the window's new sources are recorded in its prime factorization by a purity law internal to the construction.

**Theorem 7.1** (*S1 purity law*). *A new source  $q \in (n, 2n]$  divides  $\binom{2n}{n}$  exactly once: its first multiple  $2q$  already exceeds  $2n$ , so  $q$  contributes no further factor to  $(2n)!$ , while  $q > n$  keeps it out of  $(n!)^2$ . Consequently the product of all new sources of the window divides  $\binom{2n}{n}$ :*

$$\prod_{\substack{q \in (n, 2n] \\ q \text{ prime}}} q \mid \binom{2n}{n}, \quad \text{hence} \quad \prod_{\substack{q \in (n, 2n] \\ q \text{ prime}}} q \leq \binom{2n}{n}.$$

*Formalization.* `window_primes_prod_dvd_centralBinom` (`Rings.lean`), `window_primes_prod_dvd_centralBinom'`, `window_primes_prod_le_centralBinom` (`Newton.lean`); the divisibility rests on `prod_primes_dvd` and `Nat.Prime.dvd_choose_add`. The purity is the same self-containment property as `window_self_contained_dvd` / `window_composite_smooth` (the first proper multiple of an in-window source already exceeds  $2P_{\max}$ ), with the same role of 2 and the same extremal self-contained case  $P_{\min} = 2$ .

So the choice of object is forced by the structure rather than imported from history: the window's new sources sit in the prime factorization of  $\binom{2n}{n}$ . Its role in the proof is existential, closing what cannot be generated, even though its identity is purely structural. Two bounds close on it. The lower bound is pure combinatorics of the Pascal row.

**Theorem 7.2** (*Lower bound*).  $4^n \leq (2n+1)\binom{2n}{n}$ , *proved directly from  $4^n = \sum_{i=0}^{2n} \binom{2n}{i}$  with each term  $\leq \binom{2n}{n}$ .*

*Formalization.* `four_pow_le_newton` (`Newton.lean`).

The upper bound reads the same factorization in the contrapositive: an empty window would force  $\binom{2n}{n}$  to be built entirely from old sources.

**Theorem 7.3** (*Empty window  $\Rightarrow$  old sources only*). *If  $(n, 2n]$  contains no prime, then every prime factor of  $\binom{2n}{n}$  is  $\leq n$ .*

*Proof.* Any prime factor of  $\binom{2n}{n}$  divides  $(2n)!$ , hence is  $\leq 2n$ ; absent new sources in  $(n, 2n]$  it is  $\leq n$ .  $\square$

*Formalization.* `centralBinom_prime_factor_le` (`Newton.lean`).

The purity law and the two bounds above (Theorems 7.1–7.3) are proved in `Rings.lean` and `Newton.lean` and were the construction's first route to closure; they remain true and are kept in the development. The main theorem's proof term, however, closes the dense regime by a later, independent route (§8): a factorization upper bound reproved from scratch in `BinomialBound.lean`, combined with Mathlib's own central-binomial lower bound, rather than through `Newton.lean`'s in-project lower bound or `Rings.lean`'s purity-law upper bound. Theorems 7.1–7.3 are retained as an alternative derivation of the same closing idea, not because `binomial_contradiction` (§8) invokes them.

## 8. CLOSING THE EXTREMAL WINDOW: THE SELF-CONTAINED CERTIFICATE

Only the extremal window  $P_{\min} = 2$  needs closing, and only in the dense regime. We split at  $n = 512$ .

For  $2 < n < 512$  a kernel-checked decision procedure exhibits a prime in  $(n, 2n]$  directly: the range is split into finitely many blocks, and on each a witness prime is exhibited and verified by direct reduction inside Lean’s trusted kernel: no external lemma, and no appeal to compiled evaluation.

*Formalization.* `small_window_oracle`, by `decide` (the range split via `interval_cases` into blocks the kernel evaluator can handle in one call).

For  $n \geq 512$  the closure uses a separate pair of bounds from §7’s purity-law route: Mathlib’s own central-binomial lower bound, and an old-source upper bound on  $\binom{2n}{n}$  reproved independently in `BinomialBound.lean` (at most  $(2n)^{\sqrt{2n}} 4^{2n/3}$ , the same shape as Theorem 7.3 but not that theorem). Suppose  $(n, 2n]$  has no prime; combining this lower bound, this upper bound, and the size threshold below yields  $4^n < 4^n$ , a contradiction.

**Theorem 8.1** (Quantitative kernel). *For  $2 < n$ , if no prime lies in  $(n, 2n]$  then a contradiction follows. Hence every window  $(n, 2n]$  with  $n > 1$  contains a prime.*

*Proof sketch.* For  $n < 512$  use the oracle. For  $n \geq 512$ , chain the lower bound  $4^n < n \binom{2n}{n}$  (`Nat.four_pow_lt_mul_centralBinom`), the old-source upper bound  $\binom{2n}{n} \leq (2n)^{\sqrt{2n}} 4^{2n/3}$  (`window_centralBinom_le`, reproved in-project from Legendre/Kummer and primordial primitives), and the prime-free size comparison  $n (2n)^{\sqrt{2n}} 4^{2n/3} \leq 4^n$  (`threshold_inequality`) to reach  $4^n < 4^n$ .  $\square$

*Formalization.* `binomial_contradiction` (`BinomialCertificate.lean`). The upper bound (`window_centralBinom_le`, `BinomialBound.lean`) is reproved from `Choose.Factorization` and `Primorial`; the size threshold (`threshold_inequality`, `Threshold.lean`) from real convexity. Neither imports `Mathlib.NumberTheory.Bertrand`, so the main theorem’s import closure is free of it. The certificate is modular (`WindowCertificate`, `Certificate.lean`): any estimate of the same strength (a Chebyshev-type bound, or Nair’s  $\text{lcm}(1, \dots, 2n)$  argument [5]) plugs into the identical slot; a second instance, `erdos_certificate`, does so through Mathlib’s inequalities.

Non-circularity is by inspection: Mathlib’s `Nat.bertrand` and `exists_prime_lt_and_le_two_mul` are never used, and `Mathlib.NumberTheory.Bertrand` is not in the import closure of the main theorem; it appears only in `Erdos.lean`, which provides the instance-A comparison and is not imported by the main theorem.

## 9. MAIN THEOREM AND RECOVERY OF THE CLASSICAL CASE

Assembling the reduction (§6), the factorization identity (§7) and the certificate (§8) closes the atom at every scale, hence produces a prime in every window.

**Theorem 9.1** (Window non-emptiness). *For every prime  $P$  with  $2 < P$ , the window  $(P, 2P]$  contains a prime.*

*Formalization.* `structural_sieve_survivor`  $\rightarrow$  `gps_window_nonempty`  $\rightarrow$  `prime_in_window` (`GPS_StateMachine.lean`).

**Theorem 9.2** (Generalized family). *For every prime  $P_{\max}$  and every  $P_{\min} \geq 2$ , the window  $(P_{\max}, P_{\min} P_{\max}]$  contains a prime.*

*Proof.* By Theorem 4.3 it suffices to treat  $P_{\min} = 2$ , which is Theorem 9.1.  $\square$

*Formalization.* `gwindowHasVoid_of_bertrand`.

**Corollary 9.3** (Classical Bertrand–Chebyshev). *For every integer  $N > 1$  there is a prime  $p$  with  $N < p \leq 2N$ .*

*Proof.* Let  $P$  be the largest prime  $\leq N$ . If  $P = 2$  then  $N = 2$  and  $p = 3$  works. Otherwise  $2 < P$ ; Theorem 9.1 gives a prime  $q \in (P, 2P]$ , and maximality of  $P$  forces  $q > N$ , while  $P \leq N$  gives  $q \leq 2N$ .  $\square$

*Formalization.* `exists_largest_prime_le`, `bertrand_chebyshev` (`Main.lean`); the base-consistent structural form is `structural_bertrand_chebyshev`.

The constant 2 is not arbitrary: it is the smallest prime, hence the smallest admissible  $P_{\min}$ , hence the buildability edge (§3). Bertrand’s window is thus the extremal member of the family, the case  $P_{\min} = 2$ , on which all wider members depend and at which the self-containment structure is present.

## 10. REMARKS AND PROVENANCE

Comparison with prior work. The construction’s closest relatives in the literature are Jacobsthal’s function  $g(n)$ , the largest gap between consecutive integers coprime to the primorial of the first  $n$  primes [6], later sharpened by Iwaniec [7], and the generalized Bertrand-type window results of Das and Paul, who prove existence of a prime in  $[n, kn]$  for general  $k$  by direct counting and sieve estimates [8]. Both are thematically adjacent: Jacobsthal’s function bounds gaps in exactly the reduced-base setting this paper’s truncated arithmetic (§5) sieves with, and Das–Paul’s result generalizes the same window statement this paper’s family (§4) assembles. Neither, to the author’s knowledge, derives the window and its endpoint from the buildability edge of a prime base, or reduces the postulate to the single exact-count atom of §6; a search of the literature on Jacobsthal’s function, on elementary and generalized proofs of Bertrand’s postulate, and on primorial-coprimality sieve arguments did not surface a prior source assembling these three components in this form. This search is necessarily bounded: it covers indexed literature (arXiv, standard references on Jacobsthal’s function and Bertrand’s postulate) and cannot rule out unindexed material such as course notes or unpublished manuscripts, so the claim above is offered as a literature-grounded “to the author’s knowledge,” not as an assertion of absolute priority.

Machine verification. The development is checked in Lean 4 with Mathlib (toolchain pinned in the repository); there is no `sorry`, and the main theorem depends on no axiom beyond the three standard ones (`propext`, `Classical.choice`, `Quot.sound`), verified directly by `#print axioms bertrand_chebyshev`: in particular `Lean.ofReduceBool` (the axiom `native_decide` introduces) does not appear anywhere in its import closure. The small-window oracle for  $n < 512$ , the base anchors  $P \leq 83$ , and the Jacobsthal gap values are all discharged by `decide`, fully reduced and checked by Lean’s own kernel. The two widest of these (the  $n < 512$  search and the  $g(2310)$  gap check) exceed the kernel’s evaluator as a single call and are each split into blocks small enough for it to handle, rather than falling back to compiled evaluation. (A separate, Mathlib-based comparison instance in `Erdos.lean` does use `native_decide`; it lies outside the main theorem’s import closure and does not affect this.) A non-circularity audit (a search over uses) confirms that Mathlib’s own Bertrand theorem is not invoked. In addition to the full development described in this paper, the repository packages a minimal, self-contained three-theorem excerpt: the window/prime equivalence (§2), the buildability-edge result (§3), and the Bertrand–Chebyshev corollary, for independent automated verification by a third-party reviewer; that excerpt is a narrower subset chosen for statement-by-statement checkability, not a distinct or differently-scoped result from the one presented here.

Provenance. Two easy descriptions of this development are both wrong: it is not independent of Erdős, and it is not his argument under a different name. The structural reduction is independent: it derives the window and its edge from the base (§2–§3), assembles the scale-invariant family (§4–§5), and localizes the theorem to one exact-count atom whose

small, sparse and gap regimes close with no counting (§6). The atom is settled on the central binomial coefficient, whose prime factorization records the window’s content via the  $S1$  purity law (§7) and would be reached here regardless of the historical record. Erdős reached the same object in 1932 by a different route: a convergence on one mathematical object, carrying historical priority of use but no exclusivity over it. In the Lean development the two bounds are reproved in-project, so the main theorem’s import closure contains no Mathlib Bertrand theorem; the certificate is modular (§8), a second, Mathlib-based instance being retained, outside that import closure, for comparison.

Relation to classical proofs. The successive treatments of the postulate answered different questions and, accordingly, studied different objects, while converging on one theorem. Bertrand [1] posed the question of existence, read from tables of the integers; Chebyshev [2] answered it through the analytic size of  $\vartheta$  and  $\psi$ ; Erdős [4] and Nair [5] through the elementary size of  $\binom{2n}{n}$  and of  $\text{lcm}(1, \dots, 2n)$ . The present treatment asks a fourth question, the deterministic reach of a generating base, taking the base rather than the window as its primary object; the window and its endpoint are then derived, and existence of a prime becomes a corollary of the reach rather than the starting premise. At the single quantitative point all four are of the same strength, and any of the elementary certificates plugs into the same slot; what the structural layer adds is the reduction that makes one such certificate sufficient, together with the derivation of the window itself from the base.

Further directions. The finite mechanism decides every concrete scale; what a certificate supplies is the existential closure of the infinite tail. Whether that closure can be effected without a size inequality is open; every method known to the author is quantitative, of the same Chebyshev strength. Concretely, every certificate available today proceeds by contradiction: assume the window has no survivor, and derive an inequality that fails. A direct reformulation (computing the base’s own combinatorial coverage power over the window and showing it falls short of covering the window outright, rather than assuming a survivor’s absence and deriving a numerical contradiction from that assumption) would replace the existential step with a positive one, in keeping with the generative character of the rest of the construction. No such argument is known to the author at present; this is left as an open direction.

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